ETALE OFF V(I) IS EQUIVALENT TO THE RIG-ETALE CONDITION ON THE COMPLETION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof proves direction (2) by the annihilator-plus-Nakayama argument producing g in 1 + IB and direction (1) by derived completion of the naive cotangent complex quoted from lemma-NL-is-limit, with the completion flatness fact quoted from Algebra

Editorial note. The label lemma-rig-etale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the equivalent conditions of lemma-equivalent-with-artin (same chapter) for the completed algebra B^wedge
- the naive cotangent complex NL_{B/A} of a finite type algebra has finite cohomology modules, and I-adic completion is exact on finite modules (Algebra, Lemma algebra-lemma-completion-flat)
- NL^wedge_{B^wedge/A} is the completion (derived completion) of NL_{B/A} (lemma-NL-is-limit of the same chapter)
- Nakayama's lemma producing an annihilator of the form g = 1 + x with x in IB (Algebra, Lemma algebra-lemma-NAK)
- the etale criterion for ring maps via the naive cotangent complex (Algebra, Definition algebra-definition-etale)

Source reference. Original definition 00U1: sources/stacks-chow/tags.txt, source lines 1095–1095. Prerequisite pointer invoked at proof line(s) 2013; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Prerequisite pointer invoked at proof line(s) 2006; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00MB: sources/stacks-chow/tags.txt, source lines 817–817. Prerequisite pointer invoked at proof line(s) 2001; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AJS: sources/stacks-restricted/pinned-original-source.tex, source lines 596–596. Prerequisite pointer invoked at proof line(s) 2020;

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it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AJU: sources/stacks-restricted/pinned-original-source.tex, source lines 1810–1810. Prerequisite pointer invoked at proof line(s) 1987; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a Noetherian ring and let I be an ideal. Let B be a finite type A-algebra.*

(1) *If $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is étale over $\mathrm{Spec}(A) \setminus V(I)$, then $B^\wedge$ satisfies the equivalent conditions of Lemma 0AJU.*
(2) *If $B^\wedge$ satisfies the equivalent conditions of Lemma 0AJU, then there exists $g \in 1 + IB$ such that $\mathrm{Spec}(B_g)$ is étale over $\mathrm{Spec}(A) \setminus V(I)$.*

Proof. Assume $B^\wedge$ satisfies the equivalent conditions of Lemma 0AJU. The naive cotangent complex $NL_{B/A}$ is a complex of finite type B-modules and hence H^{-1} and H^0 are finite B-modules. Completion is an exact functor on finite B-modules (Algebra, Lemma 00MB) and $NL^\wedge_{B^\wedge/A}$ is the completion of the complex $NL_{B/A}$ (this is easy to see by choosing presentations). Hence the assumption implies there exists a $c \geq 0$ such that H^{-1}/I^nH^{-1} and H^0/I^nH^0 are annihilated by I^c for all n. By Nakayama's lemma (Algebra, Lemma 00DV) this means that I^cH^{-1} and I^cH^0 are annihilated by an element of the form $g = 1 + x$ with $x \in IB$. After inverting g (which does not change the quotients B/I^nB) we see that $NL_{B/A}$ has cohomology annihilated by I^c. Thus $A \to B$ is étale at any prime of B not lying over $V(I)$ by the definition of étale ring maps, see Algebra, Definition 00U1.

Conversely, assume that $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is étale over $\mathrm{Spec}(A) \setminus V(I)$. Then for every $a \in I$ there exists a $c \geq 0$ such that multiplication by a^c is zero $NL_{B/A}$. Since $NL^\wedge_{B^\wedge/A}$ is the derived completion of $NL_{B/A}$ (see Lemma 0AJS) it follows that $B^\wedge$ satisfies the equivalent conditions of Lemma 0AJU. □